

Muhammad Hamza Nadeem

AI-First Full Stack Engineer

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SUMMARY

AI-First Full Stack Engineer experienced in building production-ready AI applications using Next.js, FastAPI, PostgreSQL, and modern LLM frameworks. Skilled in RAG pipelines, agentic workflows, and multi-agent systems with LangChain and LangGraph, integrating OpenAI, Vertex AI, Bedrock, and open-source models. Proficient with Docker, Linux, CI/CD, and AWS-based deployments. Passionate about owning end-to-end development and delivering scalable AI solutions.

EXPERIENCE

AI Full Stack Engineer [Blunder Bot Technologies](#)

Jun '25 — Aug '25
Remote

- Developed and maintained full-stack web applications across frontend, backend, and database layers using modern web technologies
- Designed and implemented RESTful APIs for efficient client-server communication and scalable system architecture
- Integrated AI models into production workflows to enhance application intelligence and user experience
- Built responsive and dynamic user interfaces ensuring seamless interaction with backend services
- Contributed to CI/CD pipelines and deployment workflows, improving release efficiency and reliability
- Participated in end-to-end software development lifecycle from design and development to deployment and maintenance

MERN Engineer [Codistan Ventures](#)

May '24 — Aug '24
Islamabad, Pakistan

- Developed full-stack web applications using MERN stack (MongoDB, Express.js, React, Node.js)
- Built secure authentication and form validation systems using React Hooks and Yup validation schemas
- Designed and implemented responsive UI components using the Berry Template for improved user experience
- Developed an interactive sports gaming application focused on performance and user engagement
- Managed MongoDB database integration with backend services ensuring efficient and reliable data flow
- Collaborated with cross-functional team members to deliver scalable and production-ready web solutions

SKILLS

Relevant Skills LLMs, Workflows, DBs, Python, Computer Vision, Bedrock, React, Docker, Software Lifecycle, Vertex, Architecture, Cloud Deployment, EC2, Pipelines, AWS, PostgreSQL, LangChain, FastAPI, Tooling, CI/CD, Engineers, Vector, Linux

PROJECTS

Multimodal PDF RAG [Link](#)

- Built and evaluated a full Retrieval-Augmented Generation (RAG) benchmarking system comparing RAG Fusion, HyDE, CRAG, and GraphRAG on a real-world CRAG dataset. Designed and implemented end-to-end pipelines covering data ingestion, embedding-based indexing, retrieval, LLM-powered generation, and automated evaluation to ensure fair and reproducible comparisons across methods. Developed a Flask backend with a React (Vite) frontend to demonstrate retrieval results interactively. Delivered a modular and reusable evaluation framework with configurable pipelines and metrics, enabling systematic analysis of retrieval and generation performance for model selection and research insights.

AI Video Generation System [Link](#)

- Created animated videos typically requires separate tools for story writing, image generation, voice synthesis, scene composition, and video editing. This project automated the entire creative workflow from a single natural-language prompt, producing complete animated videos with no manual intervention.

EDUCATION

Bachelor of Science in Computer Science, FAST-NUCES

Islamabad, Pakistan

- Relevant coursework: AI, NLP, Generative AI, Agentic AI, Deep Learning.